KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 1st Term Examination, 2024
Department of Computer Science and Engineering
CSE 4111
Machine Learning

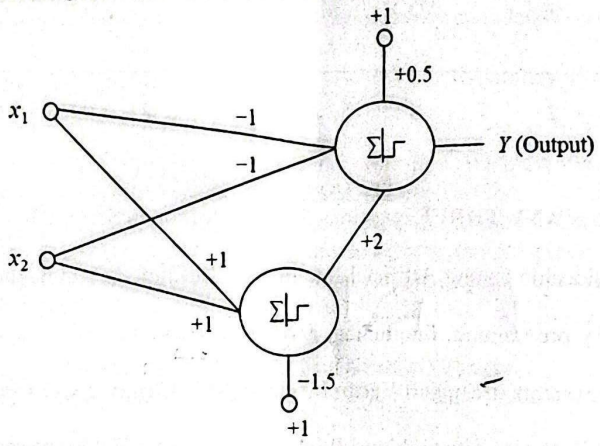
TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the immediate right column of the questions indicate full marks.
iii) The rightmost column indicates course outcomes.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) "In Machine Learning (ML), explicit programming is not performed to solve a particular problem" – justify this statement. (08) [CO1]
   b) Briefly describe the basic learning types. Arrange the learning types in order of the increasing feedback from the environment; provide justification of the ordering. (09) [CO2]
   c) Differentiate learning and generalization in the context of ML. Which one mostly indicates the performance of any ML model and what techniques are employed to achieve the indicative? (09) [CO3]
   d) Does the following neural architecture solve 2 input non-linear problem? Justify your answer considering the inputs (0,0), (0,1), (1,0), and (1,1) at x_1 and x_2. (09) [CO3]



2. a) Describe importance of bias weight value in artificial neuron construction. "Changing only the bias weight value can change the operation of a neuron" – justify the statement for appropriate logic gates. (08) [CO1]
   b) What is the condition to use a function as activation function in artificial neural network? Graphically represents two different forms sigmoidal functions and their derivatives. Explain similarity and dissimilarity between the functions. (09) [CO2]
   c) What is the local gradient in Back Propagation (BP) learning algorithm? "Local gradient may be zero even error at neural network remain" – justify the statement with an appropriate activation function. (09) [CO2]
   d) Draw diagram of RBF networks with approximate parameter values to solve AND, OR, and XOR logic operations for two inputs. (09) [CO4]

3. a) What are the various ways to produce diverse networks for ensemble construction? Which one is more effective? Why? (07) [CO1]
   b) Briefly describe independent, sequential, and simultaneous training of component networks in ensemble construction. Mention prominent ensemble methods for different ensemble training methods. (10) [CO2]
   c) Consider two computation systems: (i) multiple processing units and (ii) single high power unit. What are the preferable ensemble methods for the systems individually and why? (10) [CO4]
   d) Identify the learnable parameters for the below Convolutional Neural Network (CNN) to classify numeral images having size 28×28 pixels. Also, count the learnable parameters. (08) [CO3]

$$I_{28\times28} \rightarrow \{6K1_{5\times5}C1_{24\times24} - S_{2\times2}6S1_{12\times12}\} \rightarrow \{6K2_{5\times5}C2_{8\times8} - S_{2\times2}6S2_{4\times4}\} \rightarrow W_{o96\times10} \rightarrow O_{10}$$

where symbols hold usual meanings.

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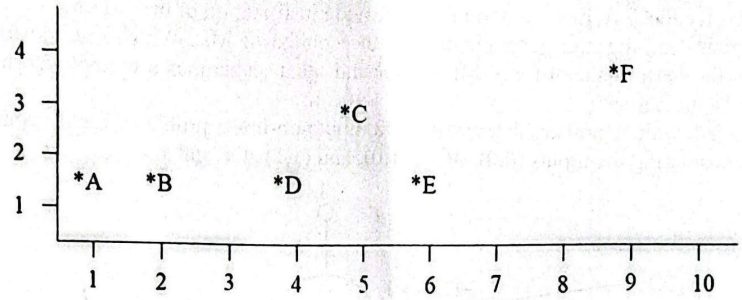
4. a) Discuss advantages and disadvantages of Decision Trees relative to Neural Networks for classification on the basis of important issues (e.g., data type, compatibilities, sample complexities, model interpretabilities). (10) [C

b) Consider the following table where attribute "Play Tennis" represents class label. Construct a Decision Tree using the table measuring information gain and then generate rules from the Decision Tree. (10) [CO3]

Sl. No.	Outlook	Temp	Humidities	Wind	Play Tennis
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Rain	Mild	High	Weak	Yes
4	Rain	Cold	Normal	Weak	Yes
5	Rain	Cold	Normal	Strong	No
6	Sunny	Mild	Normal	Strong	Yes

c) Briefly discuss strength and weakness of K-means clustering. (07) [CO2]

d) Draw a dendrogram considering six data points shown below; no need exact distance calculation. (08) [CO3]



SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) Explain the relationship among Artificial Intelligence, Machine Learning, and Deep Learning. (06) [CO1]

b) What is meant by pre-training, fine-tuning, and transfer learning? Give example of each case. (09) [CO1]

c) What are support vectors in Support Vector Machine (SVM)? How does Linear SVM work? (12) [CO2]

d) What is curse of dimensionality problem? Why is it an issue in ML? Briefly explain a dimensionality reduction technique. (08) [CO2]

6. a) Differentiate between heuristic and meta-heuristic with examples. (06) [CO1]

b) "Meta-heuristics are designed to solve a problem more quickly and efficiently than classic methods" – justify it. (10) [CO2]

c) Explain exploration and exploitation with suitable example. (10) [CO1]

d) "Crossover and mutation are not always beneficial" – is it true? If so, justify with a numeric example. (09) [CO2]

7. a) Write simple Genetic Algorithm (GA). Why is choosing proper fitness function important in GA? Explain with example. (10) [CO2]

b) Suppose, you want to minimize $f(x) = x^2$ where $-5 \leq x \leq 5$ using GA. Assume the required initial values, and consider MSB as sign bit and rest of the bits are magnitude during encoding. Now, perform calculation for one generation by hand. (12) [CO4]

c) Write an algorithm for Enhanced Edge Recombination Operator with proper example. (13) [CO3]

8. a) What is Swarm? Write properties of a Swarm Intelligence (SI) system. (05) [CO1]

b) Explain Particle Swarm Optimization (PSO) technique. What are the drawbacks of this technique? (12) [CO2]

c) Explain Ant Colony Optimization (ACO) to solve Traveling Salesman Problem (TSP). (10) [CO4]

d) Which bees perform exploration and exploitation in Artificial Bee Colony (ABC) optimization? What is significance of parameter "limit" in ABC optimization? (08) [CO2]

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